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STACK MOLD CARRIER

Abstract

A stack mold carrier for supporting a center mold section of an injection molding machine, including a platen assembly, is disclosed. The stack mold carrier includes a carrier base which includes a mold support portion for coupling with and supporting the center mold section. A linkage assembly is pre-installed on the carrier base and a first restraint system is provided for releasably retaining the linkage assembly in a first, non-operational position wherein the linkage assembly is fixed relative to the carrier base such that the stack mold carrier can be disposed on the platen assembly as an assembled unit. The stack mold carrier includes a second restraint system for releasably retaining the stack mold carrier on the platen assembly prior to connection with the center mold section.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a divisional of application of U.S. application Ser. No. 17/774,590 filed Nov. 5, 2020, which application claims the benefit of and priority to U.S. Provisional Patent Application No. 62/930,857 filed Nov. 5, 2019 under the title STACK MOLD CARRIER. The contents of both of these applications are hereby expressly incorporated by reference into the detailed description hereof.

FIELD

[0002] The present disclosure relates generally to the field of injection molding, in particular, stack mold carriers for injection molds.

BACKGROUND

[0003] It is well known in the art of injection molding to use opposing faces of adjacent mold plates, one defining a cavity, the other including a mold core which mates with the cavity, creating a cavity therebetween. These plates are brought together and molten plastic is injected between adjacent mating plates. Mold plates are cooled to cause solidification of the plastic. The plates are then separated and the finished plastic article ejected.

[0004] In order to achieve high volume production, a stack mold is often used, wherein two or more sets of molds are interspersed in sequence in which rows of mold cores and mold cavities (often referred to as mold levels or mold sets) are stacked in a single machine. In conventional stack molds, typically, a central, first mold plate is provided, for holding the mold cavities of the respective stacks of molds. Typically, the central plate comprises two or more abutting plates, and is connected to cooling lines and the like for cooling of the injected molten plastic. Two outer, second mold plates are located on either side of the central plate, with the entire system opening and closing during the molding process. One of the outer, second mold plates is affixed to a stationary platen of a conventional injection mold, while the other outer mold plate is attached to the moveable platen. The other outer, second mold plate and the central first mold plate move, relative to the stationary platen during the opening and closing process of the mold.

[0005] Stack mold carriers for supporting the center mold plate or center mold section in the injection mold and that allow the mold plate to move along the machine tie bars and/or machine linear bearing or linear guideway, or linear ways, or any other known guiding mechanisms, are known. However, stack mold carriers can be difficult, and often time consuming, to install as they often include various components that require installation on and attachment to various parts of the injection mold. Complex installation requirements associated with stack mold carriers can also impose safety risks to workers involved with the installation, repair and or operation of the mold.

[0006] Access to the mold plates of an injection mold is also required, for various reasons, such as to perform stack mold maintenance, unexpected stack mold repair, insert changes on the mold, etc. Accordingly, a stack mold carrier that can be installed on the mold in a more efficient and safe manner, and that can remain installed on the machine while the stack mold is removed and/or changed, is desirable, to not only improve safety but to also to reduce non-operational time of the mold, while installation and maintenance is in effect, which may have overall economic costs to the mold owner

SUMMARY

[0007] According to an example embodiment of the present disclosure, there is provided a stack mold carrier for supporting a center mold section of and injection mold including a platen assembly of an injection molding machine. The stack mold carrier comprises: a carrier base for coupling with and supporting the center mold section, wherein the carrier base is configured for displacement

relative to the platen assembled along a machine axis; a linkage assembly pivotally connected to the carrier base and configured for pivotally connecting to a first platen and to a second platen of the platen assembly; a first restraint system for releasably retaining the linkage assembly in a fixed position relative to the carrier base, the first restraint system defining: a first condition, wherein the linkage assembly is fixed relative to the carrier base such that pivotal movement of the linkage assembly relative to the carrier base is prevented; and a second condition wherein the linkage assembly is disposed for pivotal movement relative to the carrier base and is operably coupled to the first and second platens; and a lift support body for releasably coupling with a lifting mechanism, the lift support body supporting the stack mold carrier when subjected to a lifting force via the lifting mechanism, wherein: while the first restraint system is disposed in the first condition, the stack mold carrier is disposed for installation on the platen assembly as an assembled unit.

[0008] According to another example embodiment of the present disclosure, there is provided an injection molding machine, comprising: a machine guideway; a platen assembly mounted on the machine guideway, the platen assembly including: a first platen; a second platen; a pair of upper and lower tie bars interconnecting the first and second platens, wherein the first platen is disposed for displacement along the upper and lower tie bars relative to the second platen; a drive mechanism for displacing the first platen relative to the second platen; a stack mold carrier disposed, independently, on either side of the platen assembly for supporting a center mold section of an injection mold, each stack mold carrier comprising: a carrier base for coupling with and supporting the center mold section, wherein the carrier base is configured for displacement relative to the upper and lower tie bars along a machine axis; a linkage assembly pivotally connected to the carrier base and for pivotally connecting to the first platen and to the second platen; a first restraint system for releasably retaining the linkage assembly in a fixed position relative to the carrier base, the first restraint system defining: a first condition, wherein the linkage assembly is fixed relative to the carrier base such that pivotal movement of the linkage assembly relative to the carrier base is prevented; and a second condition wherein the linkage assembly is disposed for pivotal movement relative to the carrier base and relative to the first and second platens; wherein: while the first restraint system is disposed in the first condition, the stack mold carrier is disposed for installation on the platen assembly as an assembled unit.

[0009] According to another aspect of the present disclosure, there is provided a method for installing a stack mold carrier on an injection molding machine including a platen assembly having a machine guideway, a first platen, a second platen and upper and lower tie bars interconnecting the first and second platens, comprising the steps of: (i) providing a stack mold carrier comprising: a carrier base for coupling with and supporting a center mold section, wherein the carrier base is configured for displacement relative to the platen assembly along a machine axis; a linkage assembly including: a propeller bar pivotally connected to the carrier base; a first link arm pivotally connected to a first end of the propeller bar, the first link arm configured for operably coupling to the first platen such that the first link arm pivots relative to the first platen in response to displacement of the first platen, relative to the second platen; and a second link arm pivotally connected to a second end of the propeller bar, the second link arm configured for operably coupling to the second platen such that the second link arm pivots relative to the second platen in response to displacement of the first platen, relative to the second platen; a first restraint system for releasably retaining the propeller bar in a fixed position relative to the carrier base, the first link arm in a fixed position relative to the propeller bar, and the second link arm in a fixed position relative to the propeller bar; a lift support body for releasably coupling with a lifting mechanism, the lift support body supporting the stack mold carrier when subjected to a lifting force via the lifting mechanism; and a second restraint system for releasably coupling the stack mold carrier relative to at least the upper tie bar of the platen assembly; (ii) installing a first mounting bracket on the first platen proximal a first one of the lower tie bars and a second mounting bracket on the second platen proximal a first one of the upper tie bars that is disposed generally parallel to the

lower tie bar associated with the first mounting bracket such that the first and second mounting brackets are disposed on the same side of the platen assembly; (iii) while the first restraint system is activated such that the propeller bar is fixed relative to the carrier base, the first link arm is fixed relative to the propeller bar, and the second link arm is fixed relative to the propeller bar, releasably coupling a lifting mechanism to the lift support body of the stack mold carrier and lifting the stack mold carrier, as an assembled unit, and disposing the stack mold carrier on the machine guideway on one side of the platen assembly in close proximity to the upper and lower tie bars disposed on the one side of the platen assembly having the first and second mounting brackets; (iv) activating the carrier restraint system for releasably coupling the stack mold carrier relative to the upper tie bar of the platen assembly; (v) releasing the first link arm from the first restraint system such that it pivots relative to the propeller bar, and pivotally connecting the first link arm to the first platen via the first mounting bracket; (vi) releasing the second link arm from the first restraint system such that it pivots relative to the propeller bar, and pivotally connecting the second link arm to the second platen via the second mounting bracket; and (vii) releasing the propeller bar from the first restraint system such that it pivots relative to the carrier base.

[0010] According to another example embodiment of the present disclosure, there is provided a stack mold carrier comprising: a carrier base for releasably coupling with a center mold section of an injection mold, wherein the carrier base is configured for disposition on a machine guideway of a platen assembly of an injection molding machine and for releasable coupling relative to the platen assembly, such that the carrier base is configured for displacement, relative to the platen assembly, along the machine guideway; a linkage assembly pivotally connected to the carrier base and configured for pivotally connecting to a first platen and to a second platen of the platen assembly, wherein the linkage assembly defines: a first condition, wherein the linkage assembly is fixed relative to the carrier base such that pivotal movement of the linkage assembly relative to the carrier base is prevented; and a second condition wherein the linkage assembly is disposed for pivotal movement relative to the carrier base and is operably coupled to the first and second platens and disposed for pivotal movement relative to the first and second platens; and a tilt-impeding member coupled to the carrier base, the tilt-impeding member including an upper tie-bar engaging portion configured for disposition in opposition to and proximal at least a portion of an upper tie bar of the platen assembly; wherein: while the stack mold carrier is disposed on the machine guideway on one side of the platen assembly of the injection mold of the injection molding machine proximal corresponding upper and lower tie bars on the one side of the platen assembly, the tilt-impeding member is disposed generally parallel to and spaced apart from the corresponding upper tie bar of the platen assembly such that the upper tie-bar engaging portion is disposed in close proximity with the portion of the upper tie bar such that: the tilt-impeding member impedes displacement of the stack mold carrier about an axis that extends generally transverse to the vertical axis of the stack mold carrier such that displacement of the carrier base away from and out of engagement with the machine guideway is prevented.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] Reference will now be made, by way of example, to the accompanying drawings which show an example embodiment of the present application, and in which:

[0012] FIG. 1 is a perspective view of stack mold carrier according to an example embodiment of the present disclosure;

[0013] FIG. 1A is a perspective view of a stack mold carrier according to another example embodiment of the present disclosure;

[0014] FIG. 2 is a perspective view of an example embodiment of a platen assembly for an

injection molding machine that is configured for cooperating with the stack mold carrier of FIG. 1;

[0015] FIG. 3 is a detail view of portion of the stack mold carrier of FIG. 1;

[0016] FIG. 4 is an assembly view of the stack mold carrier of FIG. 1 being installed on the platen assembly of FIG. 2;

[0017] FIG. 5 is a perspective, front view of the stack mold carrier disposed on the platen assembly;

[0018] FIG. 5A is a detail view of the encircled area 5A of FIG. 5;

[0019] FIG. 6 is a detail rear, perspective view of the stack mold carrier and platen assembly of FIG. 5 with a carrier restraint system in place;

[0020] FIG. 7 is a detail view of the encircled area 7 of FIG. 6;

[0021] FIG. 8 is a detail view of the lift support bar of the stack mold carrier as disposed relative to the platen assembly;

[0022] FIG. 9 is a front, perspective view of a further installation step of the stack mold carrier being installed on the platen assembly;

[0023] FIG. 10 is a detail view of a portion of the stack mold carrier after removal of a portion of the propeller restraint system of the stack mold carrier;

[0024] FIG. 11 is a detail view of the completion of the installation step as illustrated in FIG. 9 of a first link arm being positioned relative to a first platen of the platen assembly;

[0025] FIG. 12 is a detail, exploded assembly view of the completion of the installation step of FIG. 11 according to an example embodiment of the present disclosure;

[0026] FIG. 13 is a front, perspective view of a further installation step of the stack mold carrier being installed relative to the platen assembly of a second link arm being positioned relative to a second platen of the platen assembly;

[0027] FIG. 14 is a front, perspective view of the stack mold carrier installed on the platen assembly;

[0028] FIG. 15 is a side view of an alternate embodiment of the stack mold carrier 10 according another example embodiment of the present disclosure;

[0029] FIG. 16 is a cross-sectional side, detail view of the stack mold carrier of FIG. 15 mounted relative to the upper tie bar of the platen assembly; and

[0030] FIG. 17 is a front view of the stack mold carrier according to the example embodiment of FIG. 15 as viewed from the linkage assembly side; and

[0031] FIG. 18 is a rear view of the stack mold carrier according to the example embodiment of FIG. 15 as viewed from the mold side or platen assembly side.

[0032] Similar reference numerals may have been used in different figures to denote similar components.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0033] Referring now to FIG. 1, there is shown a stack mold carrier 10 according to an example embodiment of the present disclosure. The stack mold carrier 10 is configured for mounting to a platen assembly 12 of an injection molding machine and is configured to support the molds, or center mold section, of the injection mold.

[0034] An example embodiment of a platen assembly 12 is shown in FIG. 2. As shown, the platen assembly 12 includes a machine guideway 14 and first and second platens 16, 18. In some embodiments, for example, the machine guideway 14 includes, but is not limited to, linear guideways for cooperating with rollers, linear guideways with bronze pads, adaptors for cooperating with linear guideways or linear bearing guideways, linear bearing guideways, linear ways, or tie bar supports. Accordingly, the platen assembly 12 may include any suitable guiding mechanism for support and guiding displacement of the moveable platens, along the machine axis, during mold opening and closing procedures. Accordingly, it will be understood that reference to machine guideway 14 is intended to include any suitable guiding mechanism known in the art for guiding displacement or reciprocating movement of the platens 16, 18 of the platen assembly 12.

[0035] Referring, in particular to the example embodiment of FIG. 2, the first platen **16** is a moveable platen while the second platen **18** is a fixed platen. In the subject example embodiment, the first and second platens **16, 18** are each mounted, independently, on sets or pairs of upper and lower tie bars **19, 21**. The first, moveable platen **16** is mounted on the upper and lower tie bars **19, 21** such that it translates along the tie bars **19, 21**, guided by the tie bars and the machine guideway **14**, relative to the second, fixed platen **18**, along the machine axis X-X, as illustrated schematically in FIG. 2. Displacement of the first, moveable platen **16** along the tie bars **19, 21** and machine guideway **14**, in a reciprocating manner relative to the second platen **18**, defines the open and closed positions of the mold (or molds).

[0036] In some embodiments, for example, a center mold section (not shown), is disposed between the first and second platens **16, 18** and is typically configured for carrying mold plates (not shown) on either side thereof. A corresponding mold plate (not shown) is mounted on or fixed or otherwise coupled to the first platen **16**, which mold plate is complementary to the corresponding mold plate disposed on the corresponding side of the center mold section. Similarly, a mold plate (not shown) is also mounted on or fixed to the second platen **18**, which mold plate is complementary to the corresponding mold plate disposed on the corresponding side of the center mold section. In some embodiments, for example, a plurality of mold plates are installed between the first and second platens **16, 18**, with each pair of complementary mold plates defining a mold cavity therebetween when the corresponding mold plates are disposed in their closed position. Accordingly, in some embodiments, the plurality of mold plates, the center mold section and the first and second platens **16, 18** are cooperatively configured to define a plurality of mold cavities between the first and second platens **16, 18**. In use, molten material is injected into the plurality of mold cavities to form molded articles, which articles are released from the mold cavities upon opening of the molds.

[0037] In general, the injection molding machine includes a driver or drive mechanism (not shown) that operates to move the first, or moveable, platen **16** relative to the second platen **18** so as to define the open and closed mold positions. In use, the first, movable platen **16** reciprocates along the fixed tie bars **19, 21**, between the open and closed positions in accordance with principles known in the art. As the first, moveable platen **16** translates between the open and closed positions to open and close the mold, the center mold section (not shown), which also carries mold plates, must also move, relative to the first and second platens **16, 18**, in order to open and close the various mold cavities defined, in part, by the center mold section.

[0038] A stack mold carrier **10** is configured to be installed on the platen assembly **12** for supporting the center mold section (or plurality of mold plates) and for co-operating with the first and second platens **16, 18** to define the one or more mold cavities. The stack mold carrier **10** serves to move the center mold section relative to the first and second platens **16, 18**, in conjunction with displacement of the first, movable platen **16** relative to the second, or fixed, platen **18** for defining the open and closed mold positions.

[0039] In the subject example embodiment, the stack mold carrier **10** includes a carrier base **100** and a linkage assembly **20**, mounted on the carrier base **100**, for operably connecting the stack mold carrier **10** to the first and second platens **16, 18** of the platen assembly **12** of the injection mold. The operable connection between the linkage assembly **20** and the first and second platens allows the stack mold carrier **10** to translate relative to the movement of the first, moveable platen **16**. In some embodiments, for example, the stack mold carrier is provided such that the linkage assembly **20** is pre-installed on the carrier base **100**, which facilitates installation of the stack mold carrier within the mold.

[0040] In some embodiments, the carrier base **100** is comprised of modular components that are cooperatively configured to define the carrier base for supporting the center mold section. In some embodiments, the stack mold carrier **10** itself is modular in that it is made up of separate components and/or systems that are cooperatively arranged to give rise to the stack mold carrier **10** described herein.

[0041] The linkage assembly **20** includes a propeller bar **22** pivotally connected to the carrier base **100**. A first link arm **24** is pivotally connected, at a first end **23** thereof, to a first end **32** of the propeller bar **22**, the first link arm **24** having a second, distal end **25** configured for pivotally connecting or coupling to the first platen **16**. In some embodiments, for example, the first link arm **24** is pivotally connected or coupled to the first platen **16** via a mounting block or bracket **1** that is fixed to the platen **16**. In other embodiments, there may be a direct connection or direct coupling between the first link arm **24** and the first platen **16**. A second link arm **26** is pivotally connected, at a first end **27** thereof, to a second end **34** of the propeller bar **22**, the second link arm **26** having a second, distal end **29** configured for pivotally connecting or coupling to the second platen **18**. In some embodiments, for example, the second link arm **26** is pivotally connected or coupled to the second platen **18** via a mounting block or mounting bracket **3** that is fixed to the platen **18**. In other embodiments, there may be a direct connection or direct coupling between the first link arm **24** and the second platen **18**. In some embodiments, for example, the linkage assembly **20** includes a harmonic arm linkage assembly.

[0042] The carrier base **100** supports the stack mold carrier **10** on the machine guideway **14** of the platen assembly **12** of the injection molding machine. In some embodiments, for example, the carrier base **100** includes rollers **31** which allow displacement of the stack mold carrier **10** along the guideway **14** of the platen assembly **12**, while in use. While rollers **31** are shown, it will be understood that the engagement between the carrier base **100** and the linear or machine guideway **14** of the platen assembly **12**, and the manner in which displacement of the carrier base **100** of the stack mold carrier **10** along the machine guideway **14** is effected, may be in accordance with any suitable means known in the art and that the stack mold carrier **10** should not be limited to rollers. In some embodiments, for example, the carrier base **100** may include bearing pads disposed for sliding movement relative to the machine guideway **14**. In some embodiments, for example, the bearing pads may include bronze, brass, or Nylon bearing pads or any low friction alloy composite. In other embodiments, for example, the carrier base **100** of the stack mold carrier **10** may include adapters for mounting to the linear guideways or linear bearing guideways instead of rollers or bearing pads for effecting engagement between the stack mold carrier **10** and the machine guideway **14**.

[0043] The stack mold carrier **10** includes a mold support portion **36** configured for receiving and/or connecting or coupling to the center mold section (not shown). In some embodiments, for example, the mold support portion **36** is mounted to the carrier base **100**. In some embodiments, for example, the mold support portion **36** is integral with the carrier base **100**. In other embodiments, for example, the carrier base **100** includes a base portion **30** and a separate mold support portion **36** that is fixed to the base portion **30** of the stack mold carrier **10**. The center mold section (not shown) may be connected to the mold support portion **36** of the stack mold carrier **10** by any suitable means known in the art such that the center mold section translates with the carrier base **100** or base portion **30** of the stack mold carrier **10**, along the machine guideway **14**, as will be described in further detail below in connection with the operation of the stack mold carrier **10** and platen assembly **12**. In some embodiments, for example, the mold support portion **36** includes a center mold section connector **37** configured for releasably connecting with the center mold section (not shown). In some embodiments, for example, the center mold section connector **37** includes receivers for coupling with corresponding threaded members (not shown) to secure the center mold section to the carrier base **100** or base portion **30** of the stack mold carrier **10**.

[0044] In some embodiments, for example, as illustrated in the example embodiment of FIG. **1**, the mold support portion **36** includes a mold support bar **38** that is mounted on and secured to the base portion **30** of the stack mold carrier **10** such that it is disposed generally opposite to the linkage assembly **20**. Accordingly, in some embodiments, for example, the mold support bar **38** is fixed to an inner surface **39** of the base portion **30**, while the linkage assembly **20** is disposed on an outer surface **43** of the base portion **30** that is generally opposite to the inner surface **39**. In the subject

example embodiment, the mold support bar **38** extends generally longitudinally between first and second ends **40**, **42** and is mounted to the base portion **30** such that it is oriented along an axis z-z that is generally perpendicular to the machine axis x-x such that the mold support bar **38** is oriented generally orthogonal relative to the machine axis x-x or machine guideway **14**. In the subject example embodiment, the mold support bar **38** and the base portion **30** are cooperatively configured such that the second end **42** of the mold support bar **38** is disposed beyond an upper end **41** of the base portion **30**. In some embodiments, the center mold section connector **37** is disposed between the first and second ends **40**, **42** of the mold support bar and is integral with the mold support bar **38** such that the mold support bar **38** and the center mold section connector **37** are of unitary one-piece construction.

[0045] In order to facilitate installation and removal of the stack mold carrier **10** on and off of the platen assembly **12** of the injection molding machine, the stack mold carrier **10** includes a propeller restraint system **46** for retaining the linkage assembly **20** in a first, non-operational, position wherein the linkage assembly **20** is fixed relative to the carrier base **100** or base portion **30**, the first, non-operational position of the linkage assembly **20**, thereby defining a first condition of the stack mold carrier **10**. Accordingly, while the linkage assembly **20** is disposed in the first, non-operational position, illustrated for example in FIGS. **1**, **4** and **5**, the propeller bar **22** is fixed relative to the carrier base **100** or base portion **30**, and the first and second link arms **24**, **26** are fixed relative to the propeller bar **22** such that pivotal movement of the propeller bar **22** relative to the base portion **30** and pivotal movement of the first and second link arms **24**, **26** relative to the propeller bar **22** is prevented, or substantially prevented.

[0046] In some embodiments, for example, the propeller restraint system **46** includes removable or releasable propeller restraints **48** for releasably retaining the propeller bar **22** in its first, non-operational position. In some embodiments, for example, the propeller restraints **48** include locking bolts that extend through corresponding bores disposed in the propeller bar **22** and base portion **30** of the stack mold carrier **10** such that the propeller bar **22** remains disposed in an upright or substantially vertical or perpendicular position, relative to the mold axis x-x. In the example embodiment, illustrated in FIG. **14**, the propeller restraints **48** include a first locking bolt **49** that extends through a corresponding bore disposed in the propeller bar **22** and base portion **30** for securing the lower portion of the propeller bar **22** against the base portion **30**, and a second locking bolt **51** that extends through a corresponding bore disposed in the propeller bar **22** and lift support body **56** of the stack mold carrier **10** for securing the upper portion of the propeller bar **22** in a fixed, upright position relative to the stack mold carrier **10**.

[0047] The propeller restraint system **46** further includes latch bars **50** for releasably retaining the first and second link arms **24**, **26**, respectively, in their fixed position relative to the propeller bar **22**. Accordingly, in some embodiments, for example, a first latch bar **50(1)** releasably retains the first link arm **24** in a fixed position relative to the propeller bar **22**, the first latch bar **50(1)** having a first end **52** releasably connected to the propeller bar **22** and a second end **53** releasably connected to the first link arm **24** such that pivotal movement of the first link arm **24** relative to the propeller bar **22** is prevented. Similarly, a second latch bar **50(2)** releasably retains the second link arm **26** in a fixed position relative to the propeller bar **22**, the second latch bar **50(2)** having a first end **52** releasably connected to the propeller bar **22** and a second end **53** releasably connected to the second link arm **26** such that pivotal movement of the second link arm **26** relative to the propeller bar **22** is prevented.

[0048] With the linkage assembly **20** disposed in its fixed, non-operational position, the stack mold carrier **10** can be lifted, as a unit, by any suitable means, for instance by means of a moveable crane arm, and disposed in position on the guideway **14** on one side of the platen assembly **12**. Similarly, a second stack mold carrier **10** can be lifted, as a unit, and disposed in position on the machine guideway **14** on the opposite side of the platen assembly **12**. The lifting of the stack mold carrier **10** as a complete unit with the linkage assembly **20** already disposed on the base portion **30**, but

restrained in its fixed, non-operational position, facilitates the installation (and/or removal) of the stack mold carrier **10** relative to the platen assembly **12** since the stack mold carrier **10** can be stored, carried and disposed relative to the platen assembly **12** as a single unit without having to separately install a base portion, a mold support and the various components of a linkage assembly, for example.

[0049] Referring now to FIGS. **1** and **3**, in some embodiments, for example, the stack mold carrier **10** includes a lift support body **56** configured for supporting the stack mold carrier **10** while the stack mold carrier **10** is subject to a lifting force, for example from a moveable crane, for displacement of the stack mold carrier **10** relative to the platen assembly **12** of the injection molding machine to facilitate installation of the stack mold carrier **10** relative to the platen assembly **12**. The lift support body **56** is cooperatively configured together with the carrier base **100**, or together with the base portion **30** and the mold support bar **38**, such that the lift support body **56** supports the weight of the stack mold carrier **10** when the stack mold carrier **10** is subject to a lifting load. Accordingly, in some embodiments, the lift support body **56** includes a lifting mechanism coupler **57** configured for coupling with the lifting mechanism to apply the lift force to the stack mold carrier. Disposition of the lift support body **56** and the lifting mechanism coupler **57** at the top end of the stack mold carrier **10** is such that the center of gravity of the stack mold carrier **10** is aligned with the lift point defined by the lifting mechanism coupler **57** where the lifting force or lifting load is applied. By having the stack mold carrier **10** configured such that the lifting mechanism coupler **57** defines a lift point at the center of gravity or aligned with the center of gravity of the stack mold carrier lift point renders the stack mold carrier **10** less prone to tipping over during installation as the lift support body **56** and lifting mechanism coupler **57** ensures plumb lifting of the stack mold carrier **10**. More specifically, by having the lift point or lifting mechanism coupler **57** configured such that it is parallel to and/or vertically aligned with the vertical axis (axis z-z) of the stack mold carrier **10** and configured in a vertical orientation is such that the assembled stack mold carrier **10** remains in a vertical orientation as it is lifted and positioned relative to the platen assembly **12**. Accordingly, the lift support body **56** and the lifting mechanism coupler **57** are engineered to achieve a plumb hoisting of the stack mold carrier **10** so that the stack mold carrier **10** can be moved and navigated, for the purpose of installation, etc., relative to the platen assembly **12** and the injection molding machine without tipping. By configuring the lift support body **56** and lifting mechanism coupler **57** to ensure that lifting of the stack mold carrier **10** occurs vertically with a reduced incidence of tipping improves the overall safety with which the stack mold carrier **10** can be moved and installed relative to the injection molding machine.

[0050] In some embodiments, for example, the lift support body **56**, includes an eyebolt **58** for receiving and releasably coupling with a corresponding hook **59** of a moveable crane arm (not shown) or other suitable equipment component for lifting and moving the stack mold carrier **10** relative to the platen assembly **12** of the injection molding machine. In some embodiments, for example, the lift support body **56** is connected to the second or upper end **42** of the mold support bar **38** such that the mold support bar **38**, the base portion **30** and the linkage assembly **20** depend from the lift support body **56**. In the subject example embodiment, the lift support body **56** is also cooperatively configured such that it cooperates with the propeller restraint system **46** for releasably retaining the propeller bar **22** in a fixed position, the lift support body **56** co-operating with the propeller restraints **48** or lock bolt for securing the propeller bar **22** in its upright, generally vertical, non-operational position.

[0051] In some embodiments, for example, the lift support body **56** is configured such that it can be brought into close proximity with the corresponding upper tie bar **19** of the platen **12** assembly as the stack mold carrier **10** is being positioned and installed relative to the injection molding machine. Accordingly, in some embodiments, the lift support body **56** is configured such that it defines a receiving portion **60** that is complementary in shape to the upper tie bar **19** such that the lift support body **56** can be disposed in close proximity to the upper tie bar when the stack mold

carrier **10** is mounted relative to the platen assembly **12**. The positioning of the lift support body **56** in close proximity relative to the corresponding upper tie bar **19** of the platen assembly **12** is illustrated, for example, in FIGS. **8** and **9**. Accordingly, in some embodiments, the receiving portion **60** of the lift support body **56** includes a concave radius. In some embodiments, the receiving portion **60** of the lift support body **56** includes a recess disposed within the lift support body **56** that is configured for receiving a corresponding portion of the upper tie bar **19** such that the lift support body **56** can be positioned in close proximity to the upper tie bar **19** to allow the stack mold carrier **10** to be positioned on the platen assembly **12** for proper operation of the mold. The lift support body **56** and the receiving portion **60** are configured such that a clearance gap **60'** is disposed between the lift support body **56** and the corresponding upper tie bar **19** when the stack mold carrier **10** is positioned relative to the platen assembly **12**. Accordingly, it will be understood that while the stack mold carrier **10** is intended to be positioned in close proximity to the platen assembly **12**, the lift support body **56** does not come into contact with the upper tie bar **19**, the receiving portion **60** and the upper tie bar **19** being cooperatively configured such that the clearance gap **60'** remains present between the lift support body **56** and the upper tie bar **19**. In some embodiments, for example, the clearance gap **60'** is a minimum of about 0.0005 inches. In some embodiments, for example, the clearance gap **60'** is about 0.010 inches. Accordingly, it will be understood that the clearance gap **60'** disposed between the receiving portion **60** of the lift support body **56** and the upper tie bar **19** is such that installation of the stack mold carrier **10** on the platen assembly **12** is free from interference between the stack mold carrier **10** and the tie bars of the platen assembly **12** that would interfere with or impede displacement of the stack mold carrier **10** along the tie bars and/or machine guideway **14** during operation of the mold. The clearance gap **60'** is also selected such that it is not so large so that the degree of displacement of the stack mold carrier **10** relative to the tie bars is such that would permit unwanted tipping of the stack mold carrier **10** relative to the tie bars of the platen assembly **12** to an extent or with such force that would increase the likelihood of damage to the platen assembly **12**, for example, and/or increase the risk of injury to persons involved in the installation of the stack mold carrier **10** and operation of the mold or injection molding machine **14**.

[0052] In some embodiments, for example, the receiving portion **60** of the lift support body **56** may include padding, for instance low friction padding, disposed at spaced apart intervals along the surface of the recessed area or receiving portion **60**. In some embodiments, the padding may prevent damage to the tie bars should the stack mold carrier **10** be disposed out of its proper position relative to the platen assembly **12** and come into contact with the tie bars. However, it will be understood that even in embodiments wherein the lift support body **56** includes padding or padding portions disposed at spaced apart intervals along the contour of the receiving portion **60**, that a clearance gap **60'** between the tie bar and the lift support body (with padded regions) remains.

[0053] As set out above, the base portion **30** supports the stack mold carrier **10** on the machine guideway **14** when the stack mold carrier **10** is installed on the platen assembly **12**. In some embodiments, in order to ensure proper positioning of the stack mold carrier **10** adjacent the corresponding lower tie bar **21**, the base portion **30** is configured such that it can be brought into close proximity with the corresponding lower tie bar **21** of the platen **12** assembly as the stack mold carrier **10** is positioned and installed relative to the injection molding machine, as shown for instance in FIGS. **11** and **14**. Accordingly, in some embodiments, the base portion **30** includes a recessed area **62** that is complimentary in shape to at least a portion of the lower tie bar **21** such that the base portion **30** can be disposed in close proximity to the lower tie bar **21** when the stack mold carrier **10** is mounted relative to the platen assembly **12**. Therefore, in some embodiments, for example, the recessed area **62** includes an area having a concave radius configured such that the base portion **30** can be disposed in close proximity to the corresponding lower tie bar **21** while maintaining a clearance gap **62'** therebetween. The complimentary shaping of portions of the stack

mold carrier **10**, relative to the shape of the upper and lower tie bars **19**, **21** of the platen assembly **12** helps to ensure that the stack mold carrier **10** can be positioned in close proximity to the platen assembly **12** to ensure proper positioning of the center mold section relative to the platen assembly **12** while still maintaining appropriate clearance gaps **60'**, **62'**, where appropriate, to help ensure proper operation of the stack mold carrier **10** and the mold. As described above in connection with the lift support body **56**, in some embodiments, for example, the clearance gap **62'** may be a minimum of about 0.0005 inches, while in other embodiments, for example, the clearance gap **62'** may be about 0.010 inches. Accordingly, it will be understood that the clearance gap **62'** is such that disposition of the stack mold carrier **10** relative to the platen assembly **12** does not bring the base portion **30** or carrier base **100** into proximity with the corresponding lower tie bar **21** so as to impede or interfere with the displacement of the stack mold carrier **10** along the tie bars and/or machine guideway **14**. As well, the clearance gap **62'** is not so large as to allow unwanted displacement or tipping of the stack mold carrier **10** relative to the lower tie bar **21** that is likely to result in damage to the platen assembly and/or increase the risk of injury to a user should the stack mold carrier **10** become dislodged from the machine guideway **14** and/or tip relative to the platen assembly **12**.

[0054] In some embodiments, for example, the stack mold carrier **10** includes a carrier restraint system or second restraint system **64** that is configured to assist with maintaining the proper positioning of the stack mold carrier **10** relative to the platen assembly **12** which may help to improve the overall safety of the stack mold carrier **10**. For example, referring in particular to FIGS. **1**, **6** and **7**, in some embodiments, in order to help ensure proper positioning of the stack mold carrier **10** relative to the platen assembly **12** such that it remains positioned relative to the platen assembly **12** and avoids tipping over towards or away from the platen assembly **12**, the carrier restraint system or second restraint system **64** is configured for retaining the stack mold carrier **10** relative to the corresponding upper tie bar **19** of the platen assembly **12**. The carrier restraint system **64**, therefore, is operable between an open condition wherein the stack mold carrier **10** is free to be positioned in close proximity to the corresponding upper tie bar **19** of the platen assembly **12** (or free to be moved away from the corresponding upper tie bar **19**), and a closed or retaining condition wherein the stack mold carrier **10** is releasably retained in position relative to the upper tie bar **19** of the platen assembly **12** such that displacement of the stack mold carrier **10** away from the corresponding upper tie bar **19** is impeded. The releasable retaining of the stack mold carrier **10** relative to the upper tie bar **19** via the carrier restraint system **64** is such that displacement of the stack mold carrier **10** away from the platen assembly **12** in a direction generally transverse to the machine axis x-x is prevented or resisted, or substantially prevented or resisted. With the carrier restraint system **64** in place and disposed in the closed or retaining condition, while displacement of the stack mold carrier **10** away from platen assembly **12** in a direction generally transverse to the machine axis x-x is prevented, or substantially prevented, displacement of the stack mold carrier **10** along the tie bars **19**, **21** and machine guideway **14** associated with the platen assembly **12**, along the machine axis x-x, with displacement of the first, moveable platen **16**, is permitted. The carrier restraint system **64**, therefore, does not impede or interfere with displacement of the stack mold carrier **10** along the tie bars **19**, **21** when disposed in the closed or retaining condition since the carrier restraint system **64** is configured such that it does not come into contact with the corresponding upper tie bar **19** unless the stack mold carrier **10** was to move out of its proper position on the machine guideway **14** putting the stack mold carrier **10** at risk of tipping or falling. It is only in such instances where the carrier restraint system **64** will actually engage the corresponding upper tie bar **19** as the contact between the components of the stack mold carrier **10** and carrier restraint system **64** will serve to prevent unwanted further displacement of the stack mold carrier **10** out of position relative to the platen assembly **12**. Accordingly, with the carrier restraint system or second restraint system **64** in place and disposed in the closed or retaining condition, tipping of the stack mold carrier **10** towards or away from the

upper tie bar **19** is prevented, or substantially prevented, which helps to improve the overall safety of the injection molding machine.

[0055] In some embodiments, for example, the carrier restraint system **64** includes a lock bar **66** that is pivotally mounted proximal the upper free end **42** of the mold support bar **38**. The lock bar **66** is mounted proximal the upper free end **42** of the mold support bar **38** such that it pivots between an open, unlocked position, as shown in FIG. **1**, wherein the lock bar **66** is disposed downwardly away from the upper tie bar **19** (illustrated schematically by directional arrow **65**), and a closed or locked position, as shown in FIG. **7**, wherein the lock bar **66** is disposed such that a tie-bar engaging portion **68** of the lock bar **66** is disposed such that at least a portion of the upper tie bar **19** is disposed between or intermediate the receiving portion **60** of the lift bar support **56** and the lock bar **66** while maintaining clearance around the upper tie bar **19**. Accordingly, it will be understood that once the lock bar **66** is secured in the closed, or locked position, the lock bar **66** and the tie-bar engaging portion **68** do not come into contact with the upper tie bar **19** during normal operation of the stack mold carrier **10** or during installation or positioning of the stack mold carrier **10** relative to the upper tie bar, the lock bar **66** and the tie-bar engaging portion **68** only coming into contact with or engaging with the upper tie bar **19** should the stack mold carrier **10** tip away from or otherwise begin to move away from the platen assembly **12**. Accordingly, once the lock bar **66** of the carrier restraint system **64** is disposed in the closed or locked condition such that the upper tie bar **19** is disposed between the recess **60** of the lift support body **56** and the tie-bar engaging portion **68** of the lock bar **66** of the carrier restraint system **64**, displacement of the stack mold carrier **10** away from the upper tie bar **19**, along an axis generally transverse to the machine axis x-x, for example axis y-y, is prevented. In some embodiments, for example, the lock bar **66** may also serve to guide the stack mold carrier **10** along the upper tie bar **19** as the stack mold carrier **10** with center mold section (not shown) is displaced along the tie bars and machine guideway **14** during molding operations. With the lock bar **66** in position and disposed in the closed or locked position, tipping or toppling of the stack mold carrier **10** away from the platen assembly **12** is prevented, or substantially prevented as the lock bar **66** would engage or contact the upper tie bar **19** preventing the stack mold carrier **10** from tipping or falling over. As well, in the event of installation errors and if the center mold section was to topple over towards the tie bars **19**, **21**, once again the lock bar **66** would hit the upper tie bar **19** which would prevent the center mold section from falling over. Therefore, the carrier restraint system **64** or lock bar **66** provides a fail-safe feature that prevents the possible toppling over of the center mold section, the carrier restraint system **64** or lock bar **66**, therefore, serving to improve the overall safety of the stack mold carrier **10** and the injection molding machine. In order to secure the lock bar **66** in its closed or locked position, locking screws **67**, or any other suitable means, are used to fix the lock bar **66** in its closed or locked position, the locking screws **67** co-operating with corresponding threaded openings or bores disposed in the mold support bar **38**.

[0056] In some embodiments, for example, in order to further secure and/or position the stack mold carrier **10** relative to the platen assembly **12**, the carrier restraint system **64** may further include a lower tie bar guide block **70** that is configured to assist with the securing and/or positioning of the stack mold carrier **10** relative to the corresponding the lower tie bar **19** such that displacement of the stack mold carrier **10** away from the lower tie bar **21**, along an axis generally transverse to the machine axis x-x, is prevented or further resisted. In some embodiments, the lower tie bar guide block **70** also serves to assist with guiding displacement of the stack mold carrier **10** along the lower tie bar **21** by preventing unwanted travel of the stack mold carrier **10** away from the tie bars **21** and/or off the machine guideway **14**. In some embodiments, for example, the lower tie bar guide block **70** is configured for removable or releasable mounting to the base portion **30** of the stack mold carrier **10**, the lower tie bar guide block **70** including a curved portion that is generally complimentary in shape to the profile of at least a portion of the lower tie bar **21**. Accordingly, in some embodiments, for example, the lower tie bar guide block **70** includes a curved portion or

recessed area **72** that is generally complimentary in shape to, the outer periphery of the lower tie bar **21**, as shown for instance in FIGS. **1** and **6**. In some embodiments, for example, the lower tie bar guide block **70** is removably or releasably mounted to the base portion **30** via bolts or threaded members that extend through the lower tie bar guide block **70** and are received within corresponding threaded openings disposed in the base portion **30**. However, it will be understood that the lower tie bar guide block **70** may be secured to the carrier base **100** or base portion **30** of the stack mold carrier **10** by any suitable means. While the lower tie bar guide block **70** is shown in the example embodiment of the stack mold carrier **10** of FIG. **1A**, it will be understood that, in other embodiments, the stack mold carrier **10** is not used in conjunction with a lower tie bar guide block **70** and that the carrier restraint system **64** may only include the lock bar **66**.

[0057] Installation of the stack mold carrier **10** on the platen assembly **12** of an injection molding machine will now be described in further detail having regard, in particular to FIGS. **1** and **4-14**.

[0058] As set out above, the stack mold carrier **10** is an assembled unit that includes a carrier base **100** (or base portion **30**), a linkage assembly **20**, and a mold support portion **36**. The first restraint system or propeller restraint system **46** ensures that the stack mold carrier **10** is disposed in a first condition wherein the linkage assembly **20** of the stack mold carrier **10** is disposed in a fixed, non-operational position where displacement of the linkage assembly components relative to the carrier base **100**, or base portion **30**, of the stack mold carrier **10** is prevented. Prior to lifting the assembled stack mold carrier **10**, the second restraint system or carrier restraint system **64** is disposed in its first, or open condition wherein the lock bar **66** is disposed in its open position wherein the lock bar **66** is disposed downwardly and away from the lift support body **56** as shown, for instance, in FIG. **1** (or FIG. **1A**), with schematic directional arrow **65**. In embodiments wherein the carrier restraint system **64** further includes a lower tie bar guide block **70**, as shown for instance in the example embodiment of FIG. **1**, prior to lifting of the assembled stack mold carrier **10**, the lower tie bar guide block **70** is temporarily placed aside or stored for later use to secure the stack mold carrier **10** in position on the platen assembly. From this position, the hook **59** of a crane arm, or any other suitable lifting means, is engages the lift support bar **56**, by way of eye bolt **58**, for lifting the stack mold carrier **10** and positioning the stack mold carrier **10** relative to the platen assembly **12** of the injection molding machine, as illustrated, for instance in FIG. **4**. As shown, the stack mold carrier **10** is lowered (see directional arrow **90** in FIG. **4**) and positioned relative to the platen assembly **12** such that the base portion **30** sits on the machine guideway **14** on one side of the platen assembly **12** in close proximity to the corresponding upper and lower tie bars **19, 21**, with the linkage assembly **20** disposed on the outside of the stack mold carrier **10**. The stack mold carrier **10** is then moved inwards, relative to the platen assembly **12**, such that the stack mold carrier **10** is positioned adjacent the corresponding tie bars **19, 21**, as illustrated by directional arrow **91** in FIG. **5** with the upper and lower tie bars **19, 21** being positioned within the corresponding recessed portion or recessed area of the lift support body **56** and base portion **30** of the carrier base **100** of the stack mold carrier **10**. In the subject example embodiment, the base portion **30** includes rollers for displacing the stack mold carrier along the machine guideway **14**, although, as set out above, the subject stack mold carrier **10** is not necessarily limited to linear guideways with rollers.

[0059] In some embodiments, the stack mold carrier **10** includes additional locating features **75** which confirm that the stack mold carrier **10** is appropriately positioned relative to the platen assembly **12**. In some embodiments, for example, the locating features **75** include a pin indicator **76** which lies flush with an indicator surface **77** once the stack mold carrier **10** is positioned in close proximity to the corresponding tie bars **19, 21**, as illustrated for example in FIG. **5A**.

[0060] Once the stack mold carrier **10** is positioned in close proximity relative to the corresponding tie bars **19, 21**, the user activates the carrier restraint system **64**, by releasing lock screws **67** from their first position (illustrated in FIG. **1**), such that the carrier restraint system **64** becomes disposed in its closed, or latched, or retaining position in order to secure the stack mold carrier **10** in position

relative to the platen assembly **12**. More specifically, the carrier restraint system **64** is activated such that the lock bar **66** associated with the upper tie bar **19** is flipped or pivoted upwards relative to the mold support bar **36** such that the upper tie bar engaging rod **68** is disposed proximal the inner side of the upper tie bar **19**, as illustrated, for example in FIG. **6**, and the lock screws **67**, that are originally used to hold the lock bar **66** in its open position are then disposed in their second position (illustrated in FIG. **6**) are used to secure the lock bar **66** in its closed position, as illustrated, for example, in FIG. **7**. In embodiments that include a lower guide block **70**, the lower tie bar guide block **70**, which was temporarily placed aside, is once again secured in position on the base portion **30** such that the lower guide block **70** is disposed in closed proximity to the corresponding lower tie bar **21**, as illustrated in FIG. **6**, with at least a portion of the lower tie bar **21** being disposed between the base portion **30** and the lower tie bar guide block **70** while maintaining a clearance gap **72'** therearound. In some embodiments, for example, the clearance gap **72'** between the lower tie bar **21** and the lower tie bar guide block **70** is similar to the clearance gap **62'** disposed between the base portion **30** and between the lower tie bar **21** and, in some embodiments may be a minimum of 0.005 inches while in other embodiments may be about 0.010 inches. Accordingly, the clearance gap **72'** is such that lower tie bar guide block **70**, when used, does not impede or interfere with displacement of the stack mold carrier along the tie bars or machine guideway **14**.

[0061] Once the carrier restraint system **64** is disposed in its closed, or latched or retaining position, such that displacement of the stack mold carrier **10** away from the platen assembly **12** along an axis generally transverse to the machine axis x-x, for instance axis y-y, is prevented, the hook **59** from the associated crane arm, or other suitable lifting means, is released from engagement with the lift support bar **56**, as illustrated in FIG. **8**, and becomes re-connected to the stack mold carrier **10**, with the hook **59** engaging the first link arm **24**, as illustrated schematically in FIG. **9** with directional arrow **92**. In some embodiments, the first link arm **24** includes an eye bolt **58** for facilitating engagement with the hook **59** of the crane arm (not shown). Only once the first link arm **24** is engaged by the hook **59** of the corresponding crane arm or lifting means, the first latch bar **50(1)** of the first or propeller restraint system **46** is removed and stored for later use. In some embodiments, the stack mold carrier **10** includes a storage area in which the latch bars **50** of the propeller restraint system **46** can be secured. An example embodiment of a latch bar secured to the stack mold carrier **10** in the storage area is illustrated in FIG. **10**.

[0062] Once the crane arm engages the first link arm **24** and the corresponding latch bar **50** is removed, the first link arm **24** is no longer fixed relative to the propeller bar **22**. The first link arm **24**, therefore, can be pivoted downwards, as illustrated by schematic directional arrow **93** in FIG. **11**, and brought into position for securing the second end **25** of the first link arm **24** to the first platen **16** or to the corresponding mounting block **1** associated with the first platen **16**. In the subject example embodiment, corresponding pairs of mounting brackets **1, 3** and **2, 4** are provided and pre-installed on the platens **16, 18** for cooperating with the link arms **24, 26** to secure the linkage assembly **20** to the platen assembly **12**. In order to ensure the first link arm **24** aligns with the corresponding mounting bracket **1** on the first platen **16**, the stack mold carrier **10** may also need to be manually jogged or urged along the machine guideway **14** during installation.

[0063] In embodiments that incorporate mounting blocks for securing the link arms to the platens of the platen assembly **12**, the mounting brackets **1, 2, 3, 4**, each include a hub portion **5** with a bore opening **6** for cooperating with the corresponding end of the first or second link arm **24, 26** and the corresponding mounting hub components **8** such that the link arms **24, 26** are pivotally connected to the respective first or second platen **16, 18** directly, or via the corresponding mounting bracket. Accordingly, in some embodiments, for example, the first and second link arms **24, 26** include openings **78** which align with the bore opening **6** in the hub portion of the corresponding mounting bracket **1, 2, 3, 4** for receiving and coupling with the mounting hub components **8** as illustrated, for example, in the exploded assembly view of FIG. **12**.

[0064] Once the first link arm **24** is pivotally connected to the corresponding mounting bracket **1** on the first platen **16**, the hook **59** associated with the crane arm, or other suitable lifting means, is released from engagement with the first link arm **24** for releasable coupling with the second link arm **26** as shown in FIG. **13** with reference to schematic directional arrow **94**. Accordingly, in some embodiments, for example, the second link arm **26** is also equipped with an eye bolt **58** for releasably engaging with the hook **59** of the associated crane arm or lifting means. Once the second link arm **26** is engaged and supported by the crane arm or lifting means, the second latch bar **50(2)** is removed and stored, in a similar manner as the first latch bar **50(1)**, and the second link arm **26** is pivoted relative to the propeller bar **22** until it becomes aligned with the hub portion **5** and bore opening **6** of the corresponding mounting bracket **3** disposed on the second platen **18**, the second link arm **26** being pivotally connected to the mounting bracket **3** in the same manner as the first link arm **24**. In order to ensure that the second link arm aligns with the corresponding mounting bracket **3** on the second platen **18**, the second platen **18** may need to be manually jogged or displaced along the tie bars **19, 21** and machine guideway **14**. Once both the first and second link arms **24, 26** are pivotally connected to the corresponding mounting bracket **1, 3** on the corresponding first or second platen **16, 18**, the propeller restraints **48** are removed thereby releasing the propeller bar **22** from its fixed position relative to the base portion **30**, the propeller bar **22** therefore being able to pivot or rotate relative to the base portion **30** of the stack mold carrier **10**. The same set-up or installation procedure is used to install a second stack mold carrier **10** on the opposite side of the platen assembly **12**.

[0065] Referring now to FIGS. **15-17** there is shown another example embodiment of a stack mold carrier **10** according to the present disclosure, wherein like reference numerals have been used to identify similar components.

[0066] In the subject example embodiment, stack mold carrier **10** is similar to the embodiment described in connection with FIGS. **1-14**, however, the subject example embodiment includes a tilt-impeding member **110** that further assists with securing and/or positioning of the stack mold carrier **10** relative to the upper tie bar **19** of the platen assembly **12**. Accordingly, the tilt impeding member **110** assists with ensuring that the stack mold carrier **10** is properly positioned relative to the platen assembly **12** of the injection molding machine.

[0067] The tilt-impeding member **110** is rigidly attached, fixed or otherwise rigidly connected to the lift support body **56** such that it extends generally perpendicular to the lift support body **56**. The tilt-impeding member **110**, therefore, extends longitudinally, generally parallel to the longitudinal axis of the upper tie bar **19** or generally parallel to the machine axis x-x, between first and second ends **112, 114**. In some embodiments, for example, the tilt-impeding member **110** includes a cross bar. In some embodiments, for example, the tilt-impeding member **110** is secured or rigidly connected to the lift support body **56** via a threaded member **119** that extends through and engages with corresponding openings **120, 121** disposed within the tilt-impeding member **110** and lift support body **56**, as illustrated, for example, in FIG. **16**. However, it will be understood that the tilt-impeding member **110** can be secured to the lift support body **56** using any suitable method known in the art to ensure a rigid connection between the two components.

[0068] At each of its first and second ends **112, 114**, the tilt-impeding member **110** includes an upper-tie-bar engaging portion **115** that is configured for disposition in close proximity to at least a portion of the upper tie bar **19** of the platen assembly **12**. Accordingly, the tilt impeding member **110** is disposed such that the upper tie bar engaging portion **115** is disposed in close proximity to the corresponding upper tie bar **19** with a corresponding clearance gap **115'** disposed therebetween. Therefore, the upper tie bar engaging portion **115** does not come into contact with and does not engage with the corresponding upper tie bar **19** unless the stack mold carrier **10** is displaced out of its proper operational position relative to the platen assembly **12**, for instance in the case of tipping or tilting of the stack mold carrier **10**.

[0069] In some embodiments, for example, each upper tie-bar engaging portion **115** includes a

friction-reducing member **116** disposed, independently, at each of the first and second ends **112**, **114** of the tilt-impeding member **110**. The friction-reducing member **116** is disposed such that it depends from an underside of the tilt-impeding member **110** and includes a recessed area **118** having a concave radius that, generally, is complimentary to at least a portion of the outer radius of the upper tie bar **19** of the platen assembly **12**. The concave radius of the recessed area **118** is such that as the stack mold carrier **10** is positioned in close proximity to the platen assembly **12** during installation of the stack mold carrier **10** on the injection mold, the tilt-impeding member **110** is disposed relative to the upper tie bar **19** of the platen assembly **12** such that the friction-reducing members **116** are disposed in direct opposition to and slightly spaced apart from the outer surface of the upper tie bar **19**. In some embodiments, for example, the friction-reducing members **116** include brass and/or Nylon bearing pads and/or low friction alloy composites that are rigidly attached, or otherwise fixed, to the tilt-impeding member **110**.

[0070] When the stack mold carrier **10** is disposed or mounted relative to the platen assembly **12** of the injection molding machine, the tilt-impeding member **110**, the friction-reducing members **116** and the upper tie bar **19** are cooperatively configured such that: (i) displacement of the stack mold carrier **10** along the upper tie bar **19**, as the carrier base **100** or base portion **30** is guided along the machine guideway **14** during mold opening and closing operations, is permitted; and (ii) rotation or tilting of the stack mold carrier **10** relative to the upper tie bar **19** is prevented, or substantially prevented. More specifically, disposition of the tilt-impeding member **110** relative to the corresponding upper tie bar **19** serves to prevent, or substantially prevent, any displacement of the stack mold carrier **10** about an axis y-y that extends generally perpendicular to a vertical axis z-z of the stack mold carrier **10** and/or that extends generally transverse to the machine axis x-x.

Therefore, if considering the machine axis x-x or the longitudinal axis of the upper tie bar as the X-axis of a Cartesian coordinate system, the vertical axis z-z of the stack mold carrier as the Z-axis of the Cartesian coordinate system and the axis y-y that extends generally transverse to the vertical axis z-z of the stack mold carrier **10**, as the Y-axis of the Cartesian coordinate system, with the X-Y-Z Cartesian coordinate system illustrated for reference in FIGS. **2** and **17**. Accordingly, the tilt-impeding member **110** prevents displacement or rotation of the stack mold carrier **110** about the axis y-y (or the Y-axis) relative to the platen assembly **12** such that one of the first and second ends **112**, **114** of the tilt-impeding member **110** becomes disposed farther away from the upper tie bar **19** than the other one of the first and second end **112**, **114**. Accordingly the tilt-impeding member **110** provides an additional safety mechanism wherein tilting or toppling over of the stack mold carrier relative to the platen assembly **12** of the injection mold, which could result in injury to a mold setter or any other individuals involved in the positioning of and placement of the stack mold carrier **10**, is prevented even once the lift support bar **56** is disengaged from the lifting device, e.g. mobile crane arm, forklift, etc. The tilt-impeding member **110** also serves to prevent and minimize the risk of toppling over even when the stack mold carrier is not physically attached to or secured to the platen assembly **12** or the injection mold by any other conventional means. For example, prior to the carrier restraint system **64** being disposed in the closed position, the tilt-impeding member **110** may serve to reduce and/or prevent toppling over of the tack mold carrier is the tilt-impeding member **110** will interfere with displacement of the stack mold carrier **10** out of contact with the guiding mechanism **14** in the absence of securing of the stack mold carrier **10** to the platen assembly **12**. In some embodiments, for example, the tilt-impeding member **110** works in conjunction with the lock bar **66**, which pivots between the open, unlocked position wherein the lock bar **66** is disposed downwardly away from the upper tie bar **19** such that it is disposed in a tie-bar-receiving position where it is ready to be disposed in close proximity to the corresponding upper tie bar **19**, and a locked position wherein the lock bar **66** pivots upwardly relative to the upper tie bar **19** such that at least a portion of the upper tie bar is disposed between the lock bar **66** and the lift support body **56**. Accordingly, in some embodiments, for example, the tilt-impeding member **110** and the lock bar **66** of the carrier restraint system **64** serve to secure and/or position

the stack mold carrier **10** to the platen assembly **12** of the injection molding machine.
[0071] While example embodiments of the stack mold carrier and restraint system have been described, it will be understood that certain adaptations and modifications of the described embodiments can be made. Therefore, the above discussed embodiments are considered to be illustrative and not restrictive.

Claims

1. A stack mold carrier comprising: a carrier base for releasably coupling with a center mold section for an injection mold, wherein the carrier base is configured for disposition on a machine guideway of a platen assembly of an injection molding machine and for releasable coupling to the platen assembly, such that the carrier base is configured for displacement, relative to the platen assembly, along the machine guideway; a linkage assembly pivotally connected to the carrier base and configured for pivotally connecting to a first platen and to a second platen of the platen assembly, wherein the linkage assembly defines: a first condition, wherein the linkage assembly is fixed relative to the carrier base such that pivotal movement of the linkage assembly relative to the carrier base is prevented; and a second condition wherein the linkage assembly is disposed for pivotal movement relative to the carrier base and is operably coupled to the first and second platens and disposed for pivotal movement relative to the first and second platens; and a tilt-impeding member coupled to the carrier base, the tilt-impeding member including an upper tie-bar engaging portion configured for disposition in opposition to at least a portion of an upper tie bar of the platen assembly; wherein: while the stack mold carrier is disposed on the machine guideway on one side of the platen assembly of the injection mold proximal corresponding upper and lower tie bars on the one side of the platen assembly, the tilt-impeding member is disposed generally parallel to and spaced apart from the corresponding upper tie bar of the platen assembly such that the upper tie-bar engaging portion is disposed in proximal to but spaced apart from the portion of the upper tie bar such that: the tilt-impeding member impedes displacement of the stack mold carrier about an axis that extends generally transverse to the vertical axis of the stack mold carrier such that displacement of the carrier base away from and out of engagement with the machine guideway is prevented.
2. The stack mold carrier as claimed in claim 1, wherein: the tilt-impeding member is an elongated member extending between first and second ends, each of the first and second ends, independently, including a friction-reducing member defining a concave radius for disposition proximal to the portion of the upper tie bar; and disposition of the stack mold carrier relative to the platen assembly such that the upper tie bar is received within the concave radius of each of the friction-reducing members of the tilt-impeding member is such that: the upper tie bar remains spaced apart from the friction-reducing members by a clearance gap; and displacement of the stack mold carrier along the upper tie bar along the machine axis is permitted while tilting of the stack mold carrier, such that the at least a portion of the carrier base is disposed upwardly away from and out of contact with the machine guideway, is prevented.
3. The stack mold carrier as claimed in claim 2, wherein: disposition of the stack mold carrier relative to the platen assembly such that the upper tie bar is received within the concave radius of the friction-reducing member of the tilt-impeding member is such that: impeding of the tilting of the stack mold carrier relative to the platen assembly is effected in the absence of releasable coupling of the carrier base to the platen assembly.
4. The stack mold carrier as claimed in claim 1, wherein: the carrier base includes: a base portion for supporting the stack mold carrier on the machine guideway; a mold support bar mounted on the base portion, the mold support bar configured for connecting to the center mold section; and a lift support body connected to the mold support bar and configured for releasably coupling with a lifting mechanism for displacement of the stack mold carrier relative to the injection molding

machine.

5. The stack mold carrier as claimed in claim 1, wherein: the linkage assembly includes: a propeller bar pivotally connected to the carrier base; a first link arm pivotally connected to a first end of the propeller bar, the first link arm configured for operably coupling to the first platen such that the first link arm pivots relative to the first platen in response to displacement of the first platen, relative to the second platen, along the machine axis; and a second link arm pivotally connected to a second end of the propeller bar, the second link arm configured for operably coupling to the second platen such that the second link arm pivots relative to the second platen in response to displacement of the first platen, relative to the second platen, along the machine axis; wherein: displacement of the first platen relative to the second platen, along the machine guideway, is such that the propeller bar pivots relative to the carrier base, the first link arm pivots relative to the first platen and relative to the propeller bar, and the second link arm pivots relative to the second platen and relative to the propeller bar.

6. The stack mold carrier as claimed in claim 5, further comprising: a first restraint system for releasably retaining the linkage assembly in a fixed position relative to the carrier base, the first restraint system defining: a first condition, wherein the linkage assembly is fixed relative to the carrier base such that pivotal movement of the linkage assembly relative to the carrier base is prevented; and a second condition wherein the linkage assembly is disposed for pivotal movement relative to the carrier base and is operably coupled to the first and second platens and disposed for pivotal movement relative to the first and second platens; wherein: while the first restraint system is disposed in the first condition, the stack mold carrier is disposed for installation on the platen assembly as an assembled unit; and a second restraint system for releasably coupling the stack mold carrier relative to an upper tie bar of the platen assembly; wherein: the releasable coupling of the stack mold carrier to the upper tie bar by the second restraint system is such that the upper tie bar remains spaced apart from the second restraint system and stack mold carrier by a clearance gap; and the releasable coupling remains effective in the absence of coupling of the carrier base with the center mold section.

7. The stack mold carrier as claimed in claim 6, wherein: the releasable coupling of the stack mold carrier relative to the upper tie bar of the platen assembly is such that displacement of the stack mold carrier away from the upper tie bar of the platen assembly, along an axis transverse, or generally transverse, to the machine axis, is impeded.

8. The stack mold carrier of claim 6, wherein: the first restraint system includes: a first latch bar having a first end releasably connected to the first link arm and a second end releasably connected to the propeller bar, the first latch bar resisting displacement of the first link arm relative to the propeller bar; a second latch bar having a first end releasably connected to the second link arm and a second end releasably connected to the propeller bar, the second latch bar resisting displacement of the second link arm relative to the propeller bar; and a propeller restraint for releasably connecting the propeller bar to the mold support portion such that rotation of the propeller bar relative to the carrier base is resisted.

9. The stack mold carrier as claimed in claim 6, wherein: the second restraint system includes: a lock bar pivotally connected to the carrier base, wherein the lock bar is operable between: a tie-bar receiving position wherein the stack mold carrier is disposed for positioning proximal a corresponding one of the pairs of upper tie bars of the platen assembly; and a locked condition wherein the lock bar is disposed for releasably retaining the corresponding upper tie bar between the lock bar and the carrier base and spaced apart therefrom; wherein: disposition of the lock bar in the locked condition is such that displacement of the stack mold carrier along the machine axis relative to the upper and lower tie bars is permitted, while displacement of the stack mold carrier away from the corresponding upper tie bar is impeded.

10. The stack mold carrier as claimed in claim 4, wherein: the stack mold carrier defines a center of gravity; and the lift support body includes a lifting mechanism coupler for releasably coupling with

the lifting mechanism; wherein: the lifting mechanism coupler is disposed on the lift support body such that the lifting mechanism coupler defines a lift point relative to the lift support body at which the lifting force to the stack mold carrier is applied, the lift point being aligned with the center of gravity of the stack mold carrier.
